

QCU Queue Analysis Report

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Executive Summary

- Average customer wait: N/A
- Estimated daily savings: ■2,667
- Utilization improvement: N/A

Segment Comparison

Time	Curr c	Curr p	Curr Wq (min)	Opt c	Opt p	Opt Wq (min)
05:00-06:00	3	41.9%	1.14	2	62.9%	8.21
06:00-07:00	5	27.7%	0.05	4	34.6%	0.27
07:00-08:00	5	31.8%	0.09	3	53.1%	2.34
08:00-09:00	5	33.7%	0.12	3	56.2%	2.87
09:00-10:00	5	36.3%	0.16	3	60.6%	3.73
10:00-11:00	5	51.7%	0.74	4	64.6%	3.03
11:00-12:00	2	95.4%	139.90	3	63.6%	5.12
12:00-13:00	3	70.7%	7.91	4	53.0%	1.51
13:00-14:00	5	48.0%	0.49	4	60.0%	2.03
14:00-15:00	5	34.3%	0.12	3	57.2%	2.87
15:00-16:00	5	29.9%	0.08	4	37.3%	0.40
16:00-17:00	5	24.7%	0.03	4	30.9%	0.17
17:00-18:00	2	57.5%	6.84	2	57.5%	6.84

Recommendations

- Remove 1 server at 05:00-06:00.
- Remove 1 server at 06:00-07:00 (waste hours: $p=0.277 \leq 30\%$). ρ becomes 0.346 (stable) and save ■84.47 in total cost.
- Remove 2 servers at 07:00-08:00.
- Remove 2 servers at 08:00-09:00.
- Remove 2 servers at 09:00-10:00.
- Remove 1 server at 10:00-11:00.
- Add 1 server at 11:00-12:00.
- Add 1 server at 12:00-13:00.
- Remove 1 server at 13:00-14:00.
- Remove 2 servers at 14:00-15:00.
- Remove 1 server at 15:00-16:00 (waste hours: $p=0.299 \leq 30\%$). ρ becomes 0.373 (stable) and save ■83.46 in total cost.
- Remove 1 server at 16:00-17:00 (waste hours: $p=0.247 \leq 30\%$). ρ becomes 0.309 (stable) and save ■85.49 in total cost.
- Estimated total daily savings: ■2,666.93.
- Average waiting time improvement: 75.0%.